Software Requirements Specification (SRS)

Smart Traffic Complaint and Monitoring System

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> **Note on derivation.** This SRS was reconstructed from the project's Software Architecture and Design Specification (SAD), Software Test Plan (STP) and UML Diagrams. Requirement IDs FR-01 to FR-15, NFR-01 to NFR-10 and SR-01 to SR-06 follow the ranges cited in the SAD and STP traceability tables. Where the source documents give a range but not the wording, the wording below is inferred from the test cases and design and is marked **[Confirm]**. FR-16 to FR-22 are new IDs for the image-upload and AI features, which the STP currently traces only as "Image upload / AI integration".

1. Introduction

1.1 Purpose

This document specifies the functional, non-functional, security and interface requirements of the Smart Traffic Complaint and Monitoring System. It is the reference against which the design (SAD) and tests (STP) are validated.

1.2 Scope

The system lets citizens report road and traffic problems and track them, and lets administrators review, prioritise, assign and resolve those complaints. A citizen may attach an optional road image, which an AI service classifies to assist the administrator. AI output is advisory only and never replaces administrator review.

Out of scope for this release (from the STP): live integration with municipal traffic authority systems, real-time traffic sensor feeds, live map integrations, and production deployment or external municipal workflow integration. Prediction accuracy is not guaranteed as a road-authority decision.

1.3 Definitions and abbreviations

Term	Meaning
Citizen	Registered end user who submits and tracks complaints
Administrator	Authorised user who manages all complaints
Complaint	A record of a reported traffic or road issue
AI prediction	Category, confidence and class probabilities returned by the ResNet18 classifier
SAD / STP	Software Architecture and Design Specification / Software Test Plan
FR / NFR / SR	Functional / Non-functional / Security requirement
MERN	MongoDB, Express.js, React, Node.js

1.4 References

- Software Architecture and Design Specification (SAD), updated 30 October 2026
- Software Test Plan (STP), updated 30 October 2026
- UML Diagrams (use-case, component and sequence diagrams)

2. Overall Description

2.1 Product perspective

The product is a web application with a three-tier core (React frontend, Node.js/Express REST API, MongoDB) plus a separate Python FastAPI inference service running a PyTorch ResNet18 model. The Express API calls the inference service only when a complaint includes an image.

2.2 User classes

User class	Description	Main capabilities
Citizen	Registered member of the public	Register, log in, submit complaint with optional image, view own complaints, track status
Administrator	Authorised staff account	View, filter, prioritise, assign and update all complaints; view dashboard; see uploaded images and AI predictions

2.3 Operating environment

- Frontend: React (Vite) in current stable Chrome and Firefox.
- Backend: Node.js with Express, MongoDB.
- AI service: Python, FastAPI, PyTorch ResNet18 with locally stored model weights.
- Local development ports: frontend 5173, Express API 5050, FastAPI 8000.

2.4 Design and implementation constraints

- MERN technology baseline with a separate Python inference service.
- Secrets and service URLs (for example `DL_API_URL`) are supplied through environment variables and never committed.
- Uploaded images are stored on the server filesystem (`server/uploads`) and served through `/uploads`; production needs persistent storage.
- CI (TrafficCare CI on GitHub Actions) runs backend Jest tests and the frontend build only.

2.5 Assumptions and dependencies

- The FastAPI service and model weights are available for AI features; if not, complaints still work.
- Users have a modern browser and network access.
- TLS, CORS policy, backup and monitoring depend on the final hosting environment (to be decided).

3. System Features and Functional Requirements

3.1 Account management

ID	Requirement	Verified by
FR-01	The system shall allow a user to register with a name, unique email and valid password, and shall reject a duplicate email with a clear message.	TC-01, TC-02
FR-02	The system shall authenticate users by email and password, deny invalid credentials without disclosing sensitive details, and support logout as designed.	TC-03, TC-04

3.2 Complaint submission and tracking (Citizen)

ID	Requirement	Verified by
FR-03	An authenticated citizen shall be able to submit a complaint with a category, description and location.	TC-05
FR-04	The system shall assign each complaint a unique ID and creation timestamp. **[Confirm]**	TC-05
FR-05	A newly created complaint shall have the status "Submitted".	TC-05
FR-06	A citizen shall be able to view a list of their own complaints.	TC-07
FR-07	A citizen shall be able to see the current status of each of their complaints.	TC-07, TC-25

3.3 Complaint management (Administrator)

ID	Requirement	Verified by
FR-08	An administrator shall be able to view all complaints.	TC-10
FR-09	An administrator shall be able to filter complaints by status and category.	TC-10
FR-10	An administrator shall be able to set a complaint's priority to an allowed value.	TC-11
FR-11	An administrator shall be able to assign a complaint.	TC-11
FR-12	An administrator shall be able to update a complaint's status to an allowed value; invalid values shall be rejected and the existing status left unchanged.	TC-12, TC-13
FR-13	An administrator shall be able to view a dashboard of aggregate complaint counts.	TC-14

3.4 Validation and access control

ID	Requirement	Verified by
FR-14	The system shall validate required fields (description, location), allowed complaint categories and allowed status and priority values, returning HTTP 400 and saving nothing when validation fails.	TC-06, TC-18, TC-19
FR-15	Administrative functions shall be available only to administrator accounts; citizens attempting them shall receive HTTP 403.	TC-09

3.5 Image upload and AI-assisted classification (extension)

These features extend the original baseline. The IDs are new and should be added to the STP traceability table.

ID	Requirement	Verified by
FR-16	A citizen shall be able to attach an optional JPG or PNG image to a complaint and preview it before submission.	TC-20
FR-17	The system shall accept only JPG/PNG files up to 10 MB and reject other uploads with a controlled error, without creating a complaint or storing a file.	TC-21, TC-22
FR-18	When a complaint includes an image, the system shall send it to the inference service for classification.	TC-23
FR-19	The system shall store the AI prediction (status, category, confidence, class probabilities) with the complaint. The status shall be one of Not Requested, Completed or Failed, and the category one of Road Damage, Normal Road or Pothole.	TC-23
FR-20	If inference fails or times out, the system shall still save the complaint and record the AI status as Failed.	TC-24
FR-21	Citizen and administrator views shall display the uploaded image (administrator) and the stored AI category and confidence accurately.	TC-25
FR-22	AI predictions shall be advisory only and shall not drive authorisation or workflow decisions.	Design review

4. Non-Functional Requirements

ID	Requirement	Verified by
NFR-01	Performance: with up to 1,000 complaints, at least 95% of list and detail requests shall complete within 3 seconds.	TC-15

ID	Requirement	Verified by
NFR-02	Usability: at least four of five first-time users shall submit a valid complaint within 3 minutes after login.	TC-17
NFR-03	Availability: target level **[Confirm, not specified in source documents]**.	Availability acceptance run
NFR-04	Compatibility: core flows shall work in current stable Chrome and Firefox.	TC-16
NFR-05	Maintainability: **[Confirm, ID is not referenced in the source documents]**. Suggested: automated backend tests and a frontend build run on every push and pull request to `main`.	TC-26
NFR-06	Reliability of error handling: failures shall produce controlled, documented error responses and never a false success.	TC-18
NFR-07	Data integrity: complaint data shall be stored consistently and reported accurately (for example dashboard counts match stored records).	TC-05, TC-14
NFR-08	Credential protection: passwords shall be hashed and never stored or returned in plaintext.	TC-01, TC-03
NFR-09	Privacy and access: users shall only access data they are authorised to see.	TC-08, TC-09
NFR-10	Robustness: malformed or invalid input shall be rejected safely without corrupting data or revealing internals.	TC-06, TC-13

5. Security Requirements

ID	Requirement	Verified by
SR-01	Citizens shall only retrieve their own complaints; requests for another citizen's complaint shall not return private data.	TC-08

ID	Requirement	Verified by
SR-02	Administrator operations shall be restricted to administrator accounts and enforced on the server.	TC-09
SR-03	Authentication shall use hashed passwords and the configured token or session mechanism; plaintext passwords shall never be stored.	TC-01, TC-03
SR-04	Status, priority and category values shall be validated against allowed lists on the server.	TC-13, TC-19
SR-05	Secrets and service configuration shall be held in environment variables and never committed to source control.	Code and repository review
SR-06	Error responses shall be controlled and shall not expose stack traces, secrets or internal details; diagnostics go to protected server logs.	TC-18
SR-07	Uploaded images and AI output shall be treated as untrusted data. File type and size shall be validated on the server. **[New, from SAD 2.6]**	TC-21, TC-22

6. External Interface Requirements

6.1 User interfaces

- **Citizen UI:** registration and login, complaint form with optional image selection and preview, complaint list and status view.
- **Administrator UI:** complaint list with filters, priority/status/assignment controls, dashboard, and display of the uploaded image and AI prediction details.

6.2 Software interfaces (REST API)

Exact routes and authorisation rules must be confirmed against the current route files **[Confirm]**.

Endpoint	Access	Purpose
POST /api/auth/register	Public	Register a user

Endpoint	Access	Purpose
POST /api/auth/login	Public	Authenticate
POST /api/auth/logout	Authenticated, if implemented	End session
POST /api/complaints	Citizen	Create complaint (multipart, optional `image`)
GET /api/complaints	Authenticated	List complaints per ownership rules
GET /api/complaints/:id	Owner or admin	Retrieve a complaint
GET /api/admin/complaints	Admin	List and filter complaints
PATCH /api/admin/complaints/:id	Admin	Update status, priority or assignment
GET /api/admin/dashboard	Admin	Aggregate counts
GET /uploads/:filename	Public static	Serve an uploaded image
GET /health (FastAPI)	Service	Model-loaded state and device
POST /predict (FastAPI)	Internal	Classify image (multipart field `file`), return category, confidence, probabilities

6.3 Error responses

Status	Use
400	Missing or invalid fields, invalid category/status/priority, rejected upload
401	Missing or invalid authentication
403	Insufficient role or permission
404	Resource not found or not available to the requester
409	Duplicate email or conflicting state

Status	Use
500	Unexpected failure, with no internal details exposed

7. Data Requirements

Entity	Fields
User	_id, name, email (unique), passwordHash, role (citizen or admin), createdAt
Complaint	_id, citizenId, category, description, locationDescription, status, priority, assignedTo, createdAt, updatedAt, imageUrl (optional), aiPrediction
aiPrediction	status (Not Requested, Completed, Failed), category, confidence, probabilities

Allowed values for complaint category, status and priority are to be listed in the final SRS **[Confirm]**. The initial status is Submitted.

8. Use Cases (summary)

Citizen: register or log in; submit complaint with optional image; view own complaints; track complaint status.

Administrator: view all complaints; filter and search; set priority and assign; update status; view dashboard summary.

See UML Figure 1 for the use-case diagram, Figure 3 for the submission and AI sequence, and Figure 4 for the status-update sequence.

9. Requirements Traceability

Requirement group	Design element (SAD)	Tests (STP)
FR-01, FR-02	Authentication middleware, MongoDB User	TC-01 to TC-04
FR-03 to FR-07	React citizen UI, Express complaint handling	TC-05 to TC-08, TC-25
FR-08 to FR-13	React admin UI, admin routes	TC-10 to TC-14
FR-14, FR-15	Validation, authorisation middleware	TC-06, TC-09, TC-18, TC-19
FR-16 to FR-22	Multer upload, FastAPI, ResNet18	TC-20 to TC-25
NFR-01 to NFR-10	See Section 4	TC-15 to TC-18, TC-26
SR-01 to SR-07	Middleware, validation, configuration	TC-08, TC-09, TC-13, TC-18 to TC-22

10. Open Items

1. Confirm wording of FR-04 and the NFR-03 availability target.
2. Define NFR-05 (the source documents cite NFR-01 to NFR-10 but never reference NFR-05).
3. List the allowed complaint categories, status values and priority values.
4. Verify endpoint paths, logout behaviour and ownership checks against the source code.
5. Add FR-16 to FR-22 and SR-07 to the STP traceability table.
6. Most manual test cases are still Not Run. Only TC-19 and TC-26 have recorded passes.
7. Confirm HTTPS, CORS, backup and persistent image storage for deployment.

11. Review and Approval

Team review and approval signatures to be added per the institution's submission process.